MESA C4 · AI AND ITS APPLICATIONS · FINAL · PROBLEM 10 · VIVA PREP

VC Pulse: your 2-hour viva guide

Everything about the project in simple words: the problem, the Nine Checks, the Components Map, every screen of the app, the AI, and the questions you'll be asked. Read it top to bottom once; then practise the 🎤 lines out loud.

Live app: vc-pulse-lime.vercel.app · **Code:** github.com/nawazalam4u/vc-pulse · **Video:** Downloads/VC_Pulse_Demo_Video.mp4 (3:24)

How to use your 2 hours

Time	Do this
0:00-0:15	Watch the 3-minute video once. Read section 1 (the problem) and memorise the 60-second pitch below.
0:15-0:50	Sections 2 and 3: the **Nine Checks** and the **Components Map** (the professor's focus). Say each 🎤 line out loud.
0:50-1:20	Section 4: open the live app next to this guide and click through every screen as you read.
1:20-2:00	Sections 5-8: the score, safety, AI words, then practise the questions in section 9 with a friend.

🎤 **Say this:** VC Pulse is an AI assistant for Vikram, a founder raising money for his logistics software company. He tracks 50 investors. Twice a week, it reads live news, blogs and podcasts from the last 30 days, uses Gemini Flash to decide what each signal means for his company and why, and gives every investor a score and an action: reach out, nudge, warm, wait or park. It only updates four columns in his table; his own notes are protected by the database itself, and he sends every email himself.

1 · The problem, in plain words

Imagine you want 50 neighbours to join your football team. Knocking on every door every day is exhausting. But if Rahul just posted "I want to play football this weekend", that's the door to knock on today. If Priya has been away for three months, skip her this week. **VC Pulse is the friend who reads the neighbourhood news twice a week and tells you which doors to knock on, and why**, without ever writing in your own notebook.

The real version: Vikram is raising a **Seed round** (his first big investment) for **Haulr**, a software company for trucking and shipping businesses in India. He tracks 50 **VCs** (venture capitalists: people who invest in startups) in a spreadsheet: name, firm, what they invest in, when he last spoke to them, whether he's waiting for a reply, and his private notes. The sheet tells him **who**, but not **who is worth contacting right now**.

What the brief and the professor asked for (5 requirements)

#	Requirement	How VC Pulse does it	Where to show it
1	**Real information** from the last 30 days, not invented	Live Google News, the investors' own blogs, Apple Podcasts; every item keeps its real link and date	"Latest live evidence" feed; evidence in any card

2	**Gemini Flash explains why** a signal matters	💡 "Why it matters for Haulr" on cards and in the drawer, with honest caveats	Any card's yellow box
3	**Useful ranking** from evidence, not random	Score 0-100 from what each signal means, fit and timing; maths shown line by line	"Why NN/100" in the drawer
4	**Safe updates:** only 4 signal columns change	The database itself blocks the robot from touching notes; proof on every run	"Notes guard" tab → Attack button
5	**Working app** + GitHub + Components Map	Live on Vercel, public repo, swimlane map	The link; "How it works" → Downloads

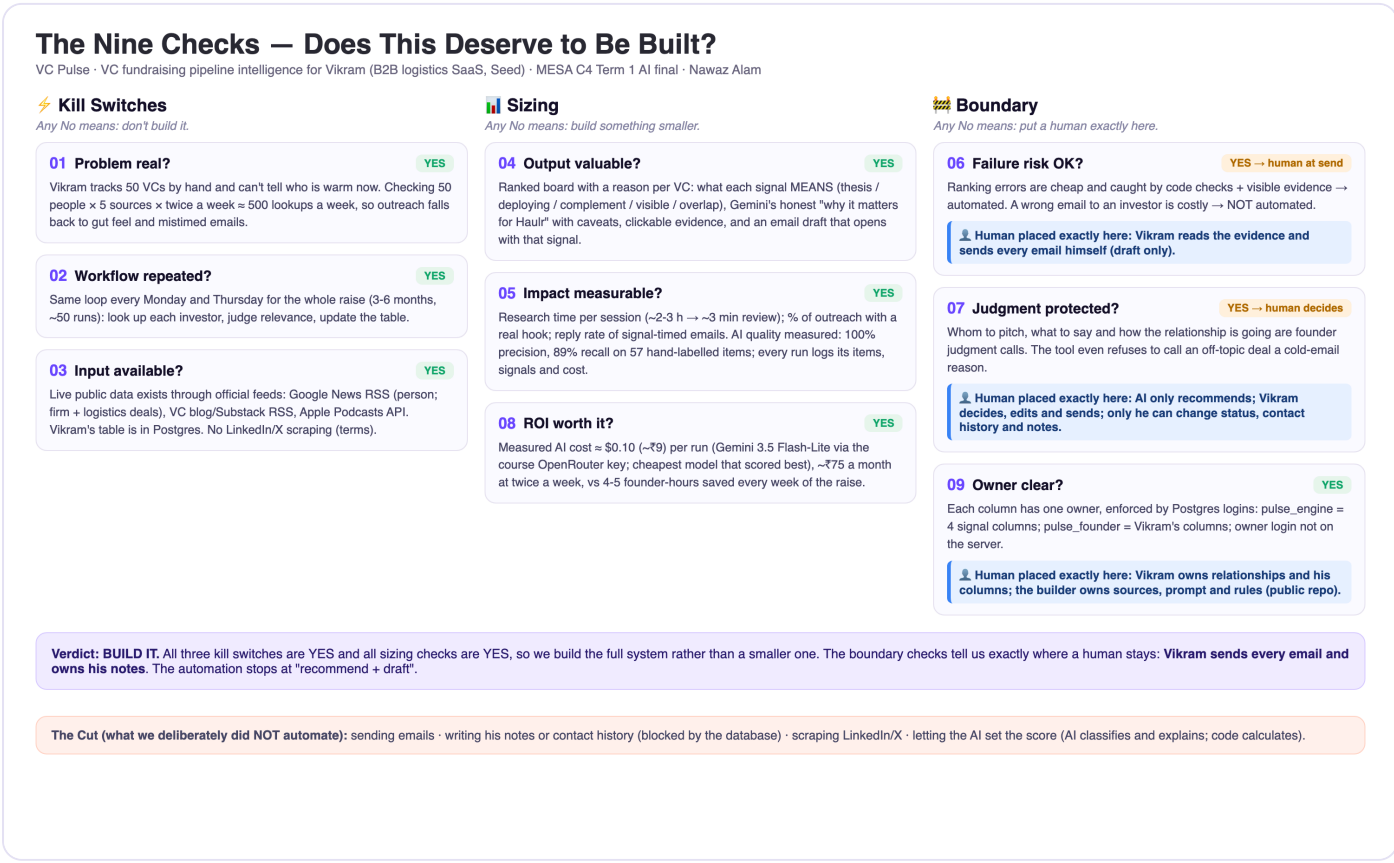
Options we considered (and why we chose ours)

Option	Why not
Google Alerts + spreadsheet	Still manual, no ranking, nothing updates the table
Zapier / no-code	Hard to guarantee "only 4 columns"; row updates can overwrite notes
Scrape LinkedIn / X	Against their rules, accounts get banned
Just ask the AI to search	Can invent facts and dates; scores change every run; not explainable
Ours: real feeds → Gemini judges → code checks and scores → database guards notes	Live, legal, cheap, every claim has a link, same evidence = same score, the key rule enforced by the database

2 · The Nine Checks: "Does this deserve to be built?"

The Nine Checks are the course's way to decide **before building** whether an AI tool is worth it. They come in three groups, and each group has a rule for what a "No" means:

Group	Checks	If any answer is NO…	In simple words
⚡ **Kill switches**	01, 02, 03	don't build it	Is there even a real, repeating problem with data to work on?
📊 **Sizing**	04, 05, 08	build something smaller	Is it worth building, and can we prove it helps?
🚧 **Boundary**	06, 07, 09	put a human exactly here	Where must a person stay in charge?



01 Problem real? YES

Does Vikram really have this pain? **Yes:** timing decides whether an investor reads your email, and checking 50 people in 5 places twice a week is about 500 lookups a week, so founders end up guessing. *Example:* an investor who wrote about logistics last week will read an email about logistics this week.

> 🎤 **Say this:** The problem is real because outreach timing matters and the manual work is too big to do consistently.

02 Workflow repeated? YES

Does it happen again and again? **Yes:** every Monday and Thursday, for the whole 3-6-month raise, the same steps: look up each investor, judge relevance, update the sheet. Repetition is what makes automation worth it.

03 **Input available?** YES

Can the AI get the data? **Yes:** public, official feeds (Google News RSS, blog/Substack RSS, the Apple Podcasts search API) plus Vikram's table in the database. **Not** LinkedIn or X scraping (against their rules).

🎤 **Say this:** All three kill switches are YES, so the project deserves to exist.

04 **Output valuable?** YES

Is what we produce useful? A ranked board, a reason per investor, Gemini's "why it matters", clickable evidence, and an email draft that opens with the signal. Vikram can act in minutes.

05 **Impact measurable?** YES

Can we prove it works? Time per session (~2-3 hours → ~3 minutes), % of emails with a real reason, reply rates, and AI accuracy we measured: **100% precision** (never a false claim) and 85-89% recall on 57 hand-checked items. Every run logs items, signals and cost.

08 **ROI worth it?** YES

Is the benefit bigger than the cost? About **$0.10 (≈ ₹9) per run**, ≈ ₹75 a month, versus 4-5 hours of founder time saved every week.

🎤 **Say this:** All sizing checks are YES, so we built the full system, not a smaller one.

06 **Failure risk OK?** YES → human at send

What if it's wrong? A wrong *ranking* is cheap (Vikram sees the evidence) and is reduced by code checks. A wrong *email* to an investor is costly, so **sending is never automated**. Human placed at: reading the evidence and pressing send.

07 **Judgment protected?** YES → human decides

Who makes the judgment calls? Whom to pitch, what to say and how the relationship is going belong to Vikram. The AI only recommends; it even refuses to call an off-topic deal a reason for a cold email.

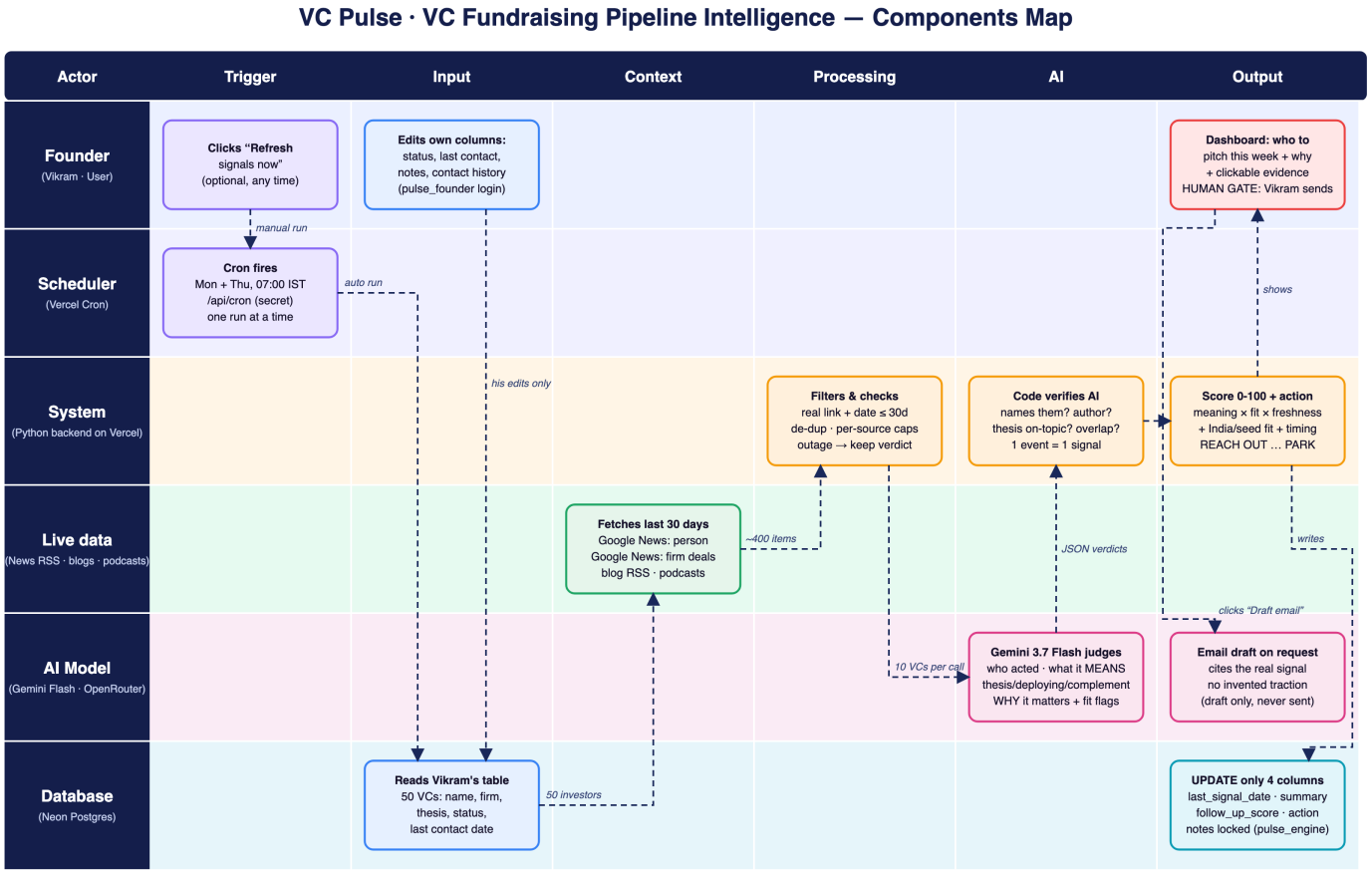
09 **Owner clear?** YES

Who owns what? Each column has one owner, enforced by two database logins: the robot owns 4 signal columns; Vikram owns his columns; the builder owns the code.

🎤 **Say this:** The boundary checks put a human exactly where judgment matters: Vikram decides and sends every email.

Verdict: build it. The Cut (what we deliberately did *not* automate): sending emails · writing his notes or contact history · scraping LinkedIn/X · letting the AI set the score (AI explains, code calculates).

3 · The Components Map: how the parts talk to each other



How to read it

Rows = the actors (who does the work): Vikram, the scheduler (Vercel Cron), our backend (Python on Vercel), the live data sources, the AI (Gemini Flash), and the database (Neon Postgres). **Columns = the stages**, left to right: **Trigger** (what starts it) → **Input** (what we start from) → **Context** (outside information) → **Processing** (cleaning) → **AI** (judgment) → **Output** (what changes). **Dashed arrows** show data moving.

Follow one run, box by box (with a real example)

#	Box	What happens	Example: Avaana Capital
1	Trigger: Cron / Refresh	Every Monday and Thursday at 7 AM, Vercel's scheduler wakes the app (or Vikram presses Refresh). Only one run at a time.	Thursday 07:00: run starts
2	Input: Vikram's table	Read the 50 investors: name, firm, thesis, status, last contact.	Anjali Bansal · Avaana Capital · Mumbai · not contacted yet
3	Context: live data	For each investor, fetch the last 30 days from Google News (person, firm + logistics), their blog feed and podcasts (~400 items).	"Drivn raises Rs 45 crore in seed funding from Avaana"
4	Processing: filters	Keep only items with a real link and a date within 30 days; remove duplicates; each source keeps its share. If a source fails, keep last run's verdict.	Item kept: real link, dated 22 Sep

5	AI: Gemini judges	Gemini Flash reads ~10 investors per call and labels each item: is it really them? what does it **mean** (thesis / deploying / complement / visible / overlap)? how close to logistics? Then writes "why it matters" and flags geography and stage.	firm · invested · 🚀 deploying · fit 2/3 · India match · seed match
6	Code verifies AI	Our code double-checks Gemini: the text must name the investor; a deal can't be a "thesis"; a stake sale isn't deploying; no company names from memory; one event counts once.	Avaana named in the headline ✓
7	Score + action (code)	Plain arithmetic: meaning × topic fit × who acted × freshness, plus fit and timing → a score and an action.	75 → WARM (see section 5)
8	Output: update 4 columns	Write only last_signal_date, signal_summary, follow_up_score, recommended_action, using the robot login, which **cannot** touch notes.	Her notes "Ask Meera for an intro" unchanged
9	Output: dashboard	Vikram sees the board, clicks the evidence, asks for a draft.	Card shows 75, the reason and the evidence
10	Human gate	Vikram decides, edits and sends the email himself (Gemini drafts on request).	He chooses whether to write this week

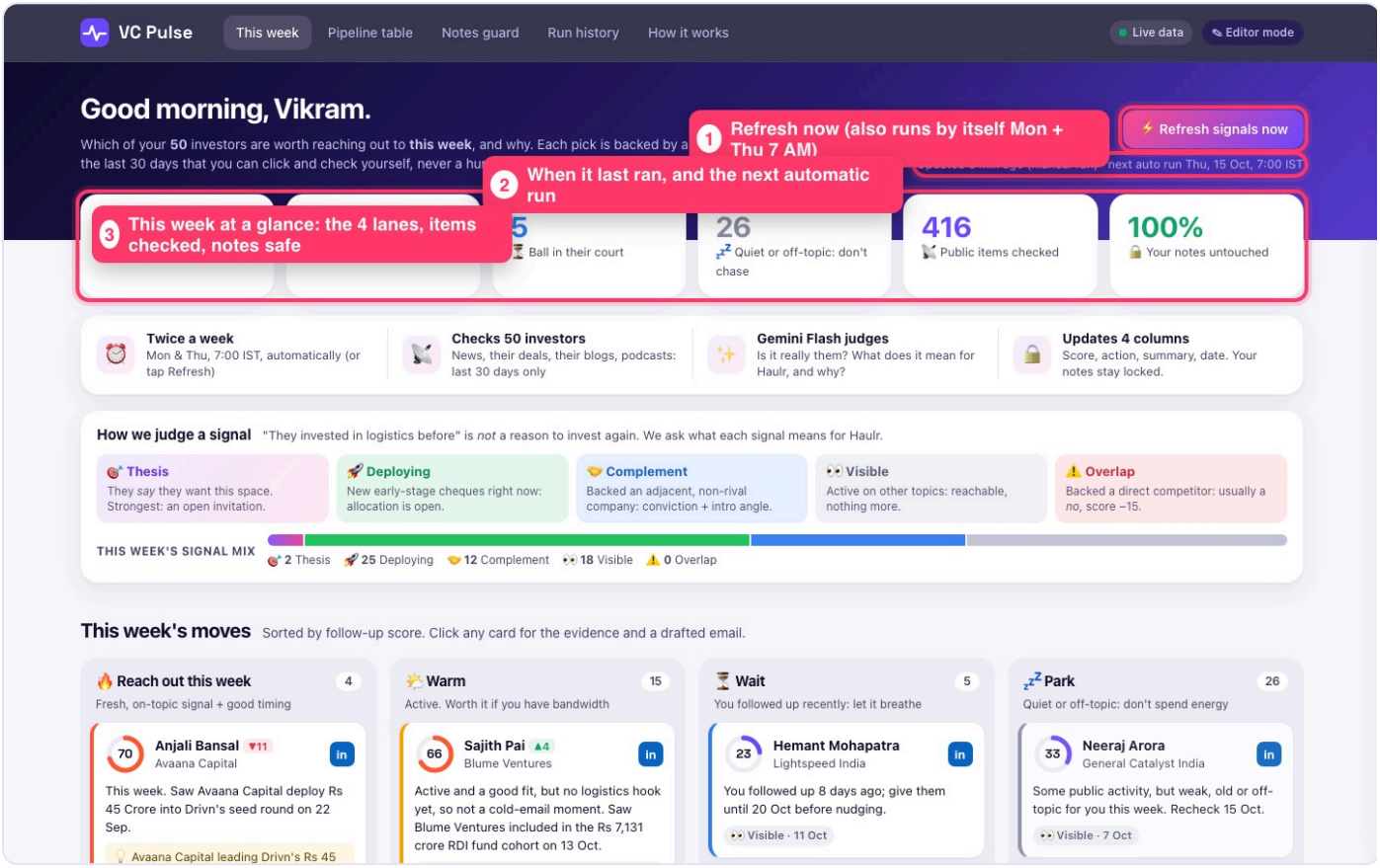
🎤 **Say this:** The map shows a pipeline: a trigger starts it, we combine Vikram's table with live data, Gemini judges meaning, our code checks the AI and does the maths, only four columns are written, and a human makes the final call.

Why is each part where it is?

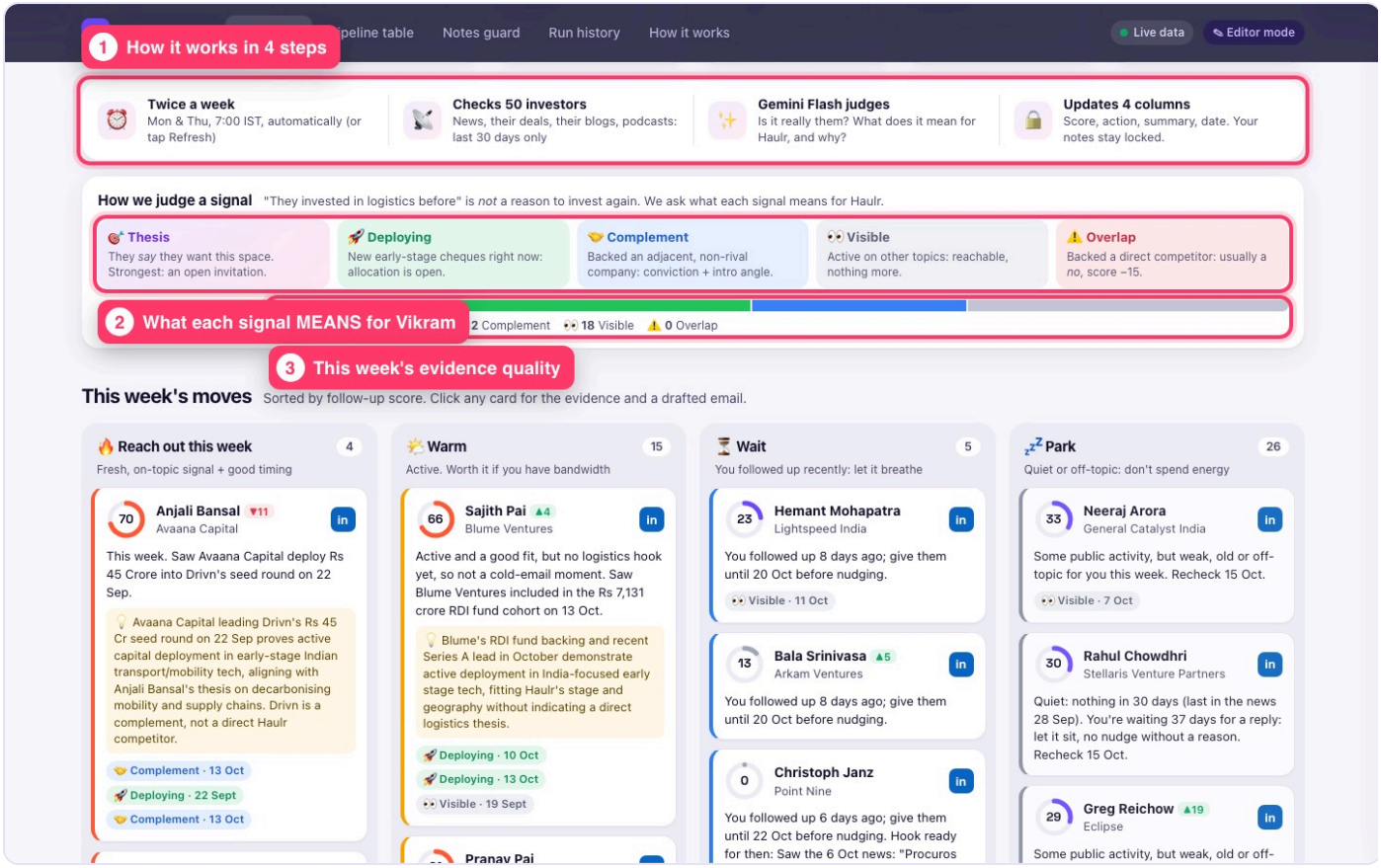
- **Why is "Code verifies AI" after Gemini?** AI can be wrong or overconfident; code checks are cheap insurance (that's how we reached 100% precision).
- **Why is the score done by code, not AI?** So the same evidence always gives the same score and every point can be explained.
- **Why does the database row write only 4 columns?** The brief's critical rule; the robot's login physically can't write anything else.
- **Why is the human gate at the end?** Boundary checks 06 and 07: sending and judging stay with Vikram.
- **Why a scheduler?** The brief says "twice a week, before he starts work", so it runs without anyone pressing a button.

4 · A guided tour of the app (open it alongside)

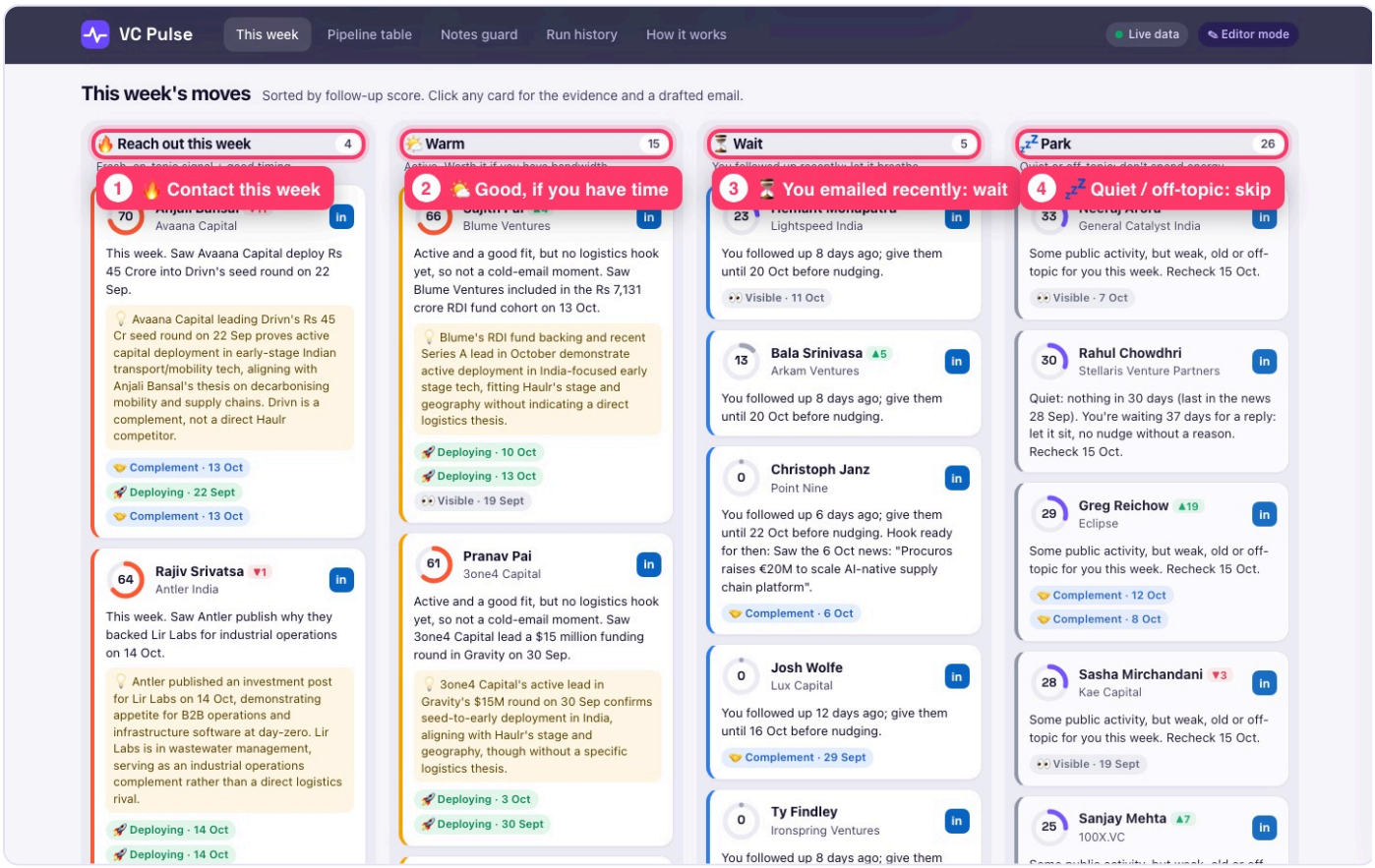
The numbered red boxes match the video. Five tabs at the top: **This week · Pipeline table · Notes guard · Run history · How it works**. *Screenshots are from 15 Oct, 3:40 AM; scores and lanes change a little after every run (new news arrives, and Gemini reads headlines slightly differently each time), so your screen will show different names and numbers. The layout and meaning stay the same.*



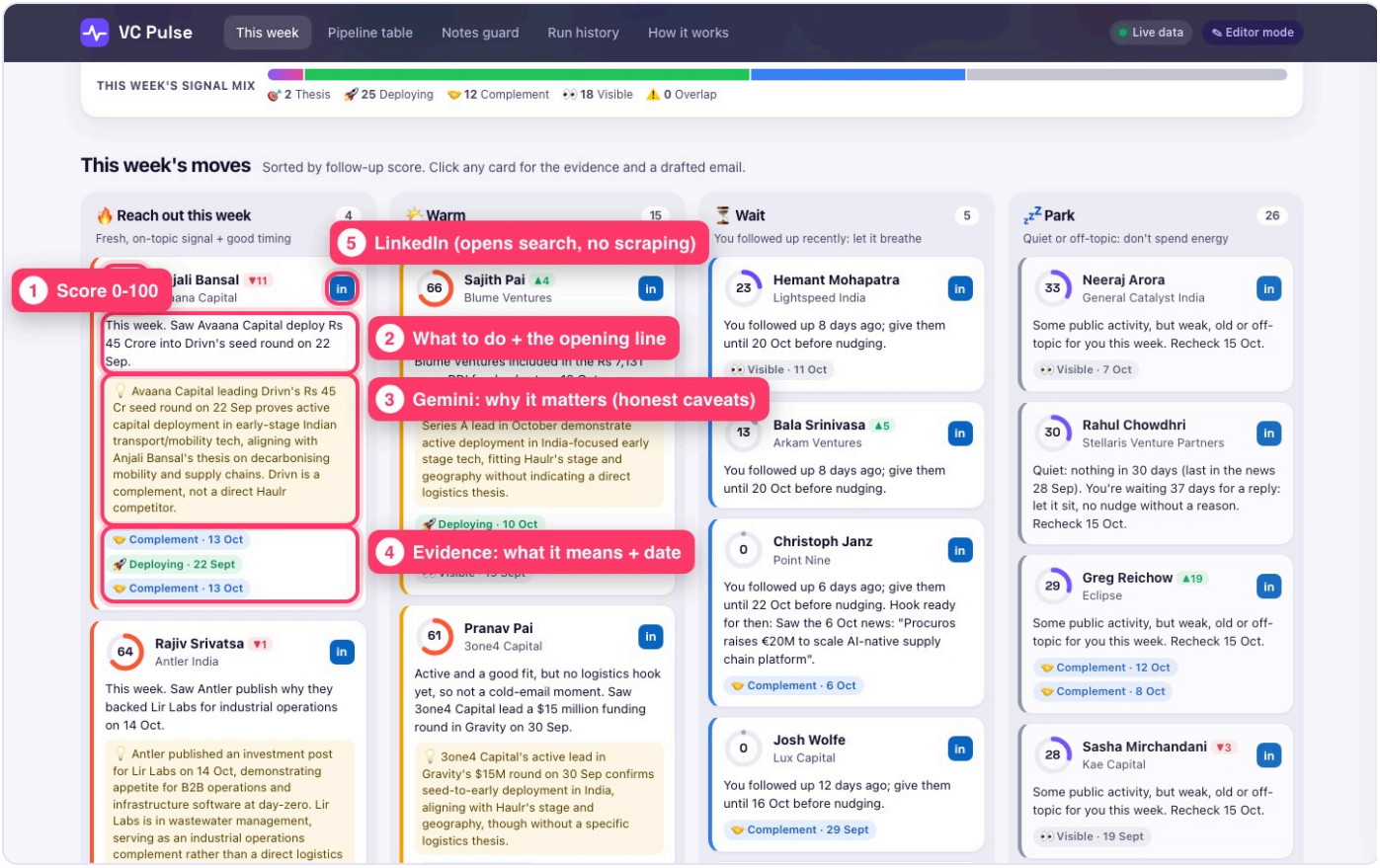
Top of "This week". 1 Refresh button (the app also runs by itself Mon + Thu 7 AM). 2 When it last ran and when it runs next. 3 The week at a glance: how many to contact, warm, wait, quiet; how many public items were checked; "100% notes untouched" is the safety proof.



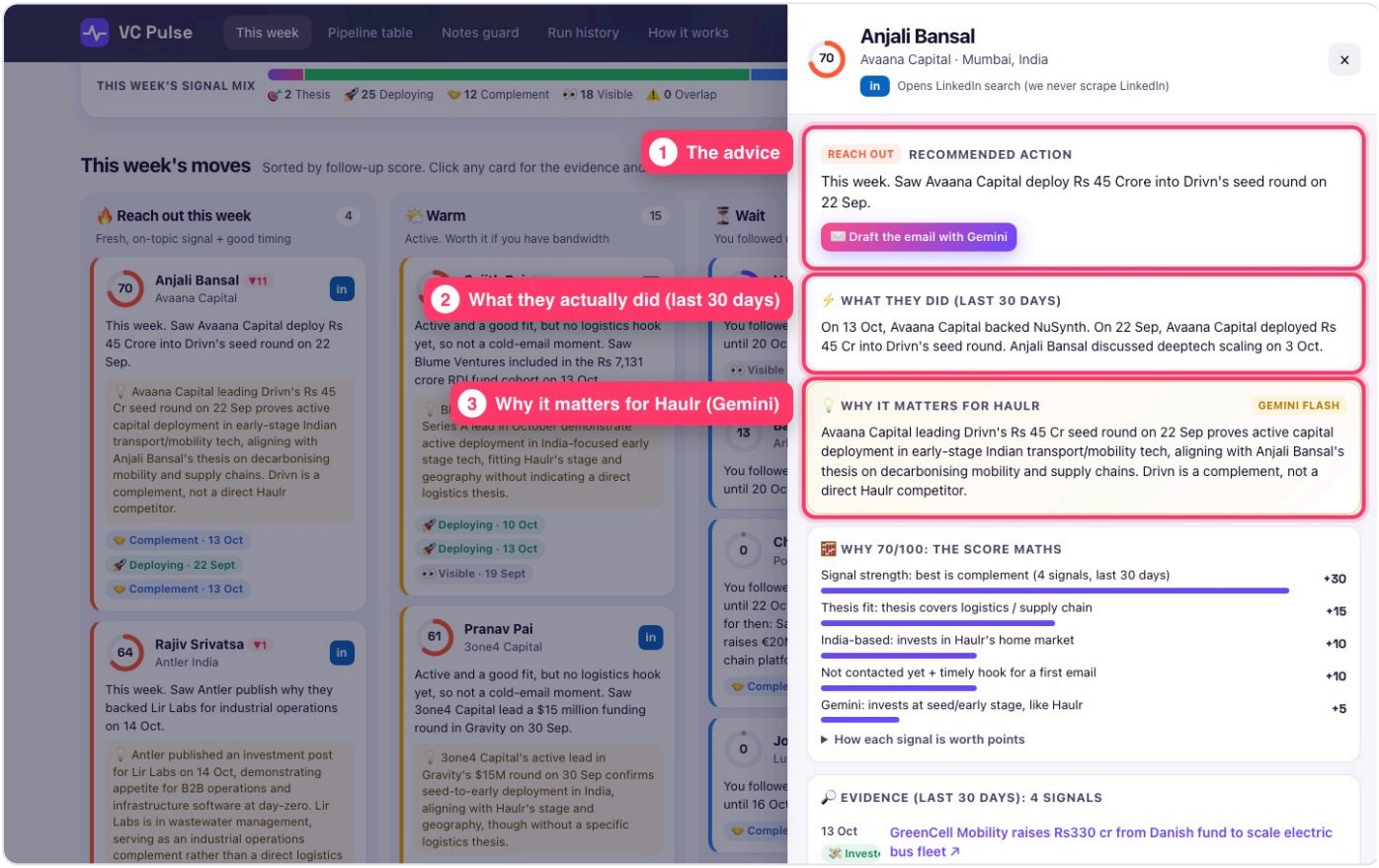
1 How it works in 4 steps. 2 **How we judge a signal**: the five meanings (the professor's key question). 3 This week's signal mix: how much of the evidence is strong (thesis) vs weak (visible).



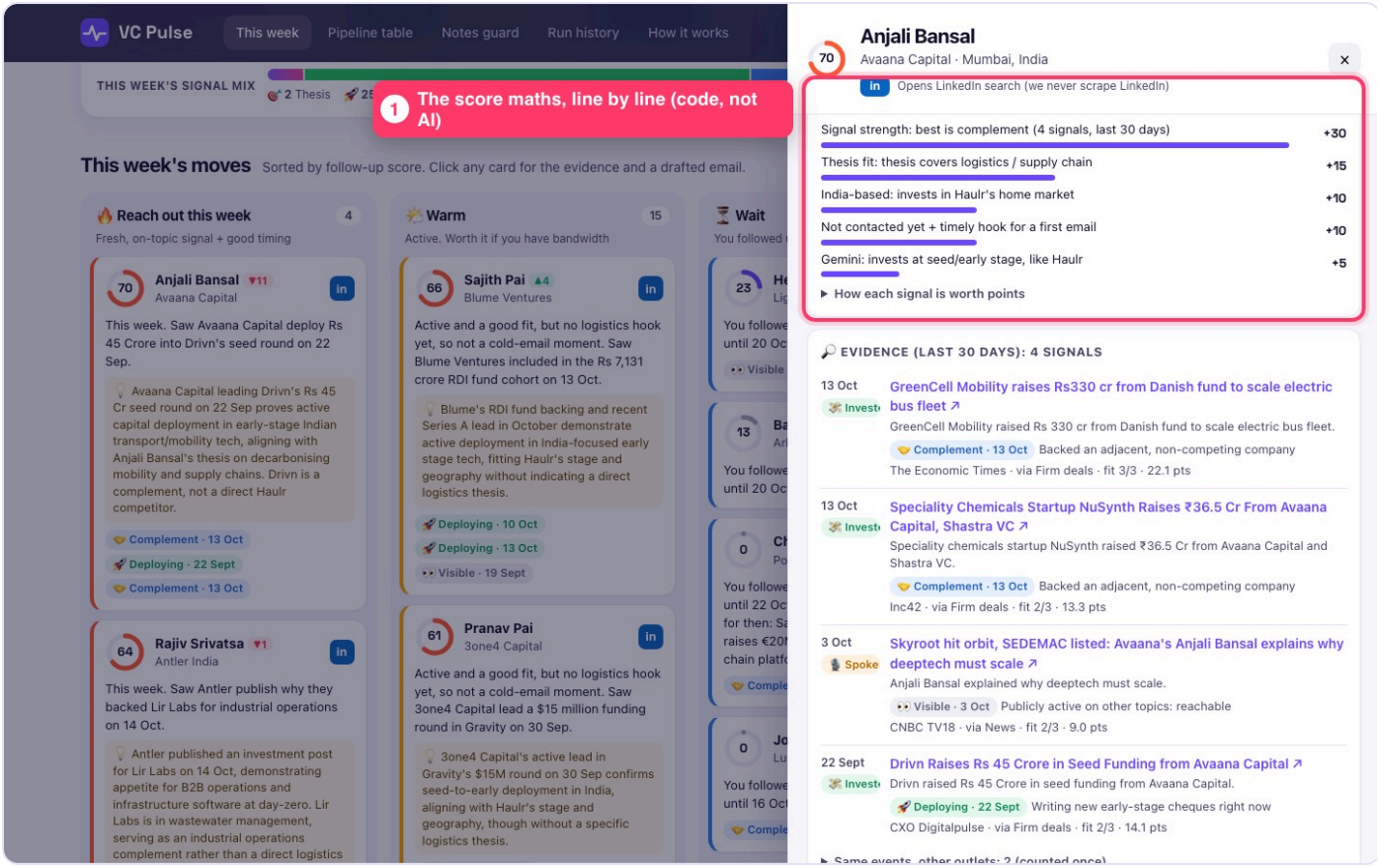
The board: four lanes. 1 🔥 Contact this week (REACH OUT for new investors, NUDGE for ones he's waiting on). 2 🌤️ Warm: worth it if he has time. 3 ⏳ Wait: he emailed less than 14 days ago, so don't pester. 4 💤 Park: quiet or off-topic, so skip this week.



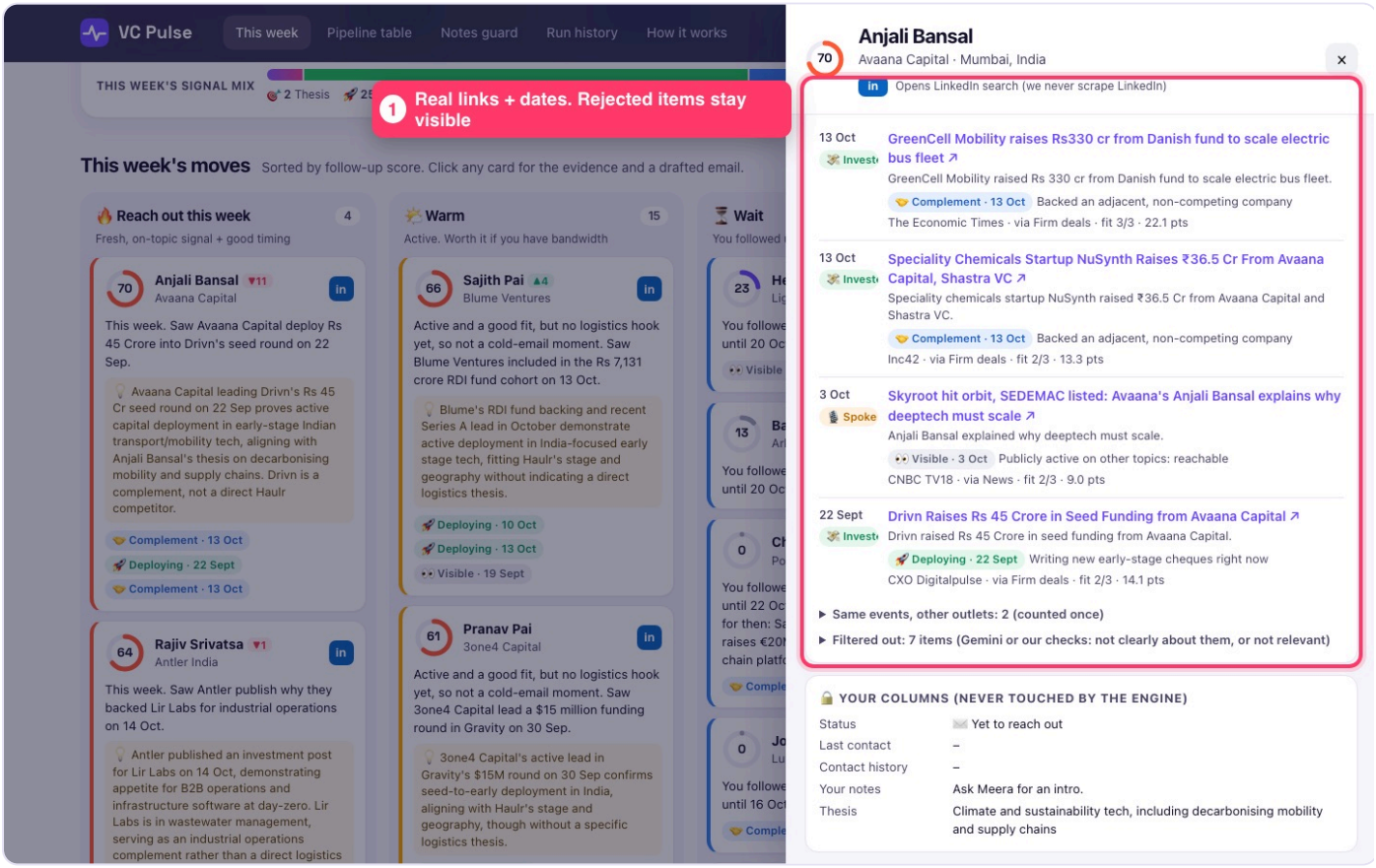
One card. 1 Score out of 100 (▲/▼ shows change since last run). 2 The advice and the opening line for an email. 3 💡 Gemini's reason, including what it doesn't prove. 4 Evidence chips: the meaning and the date. 5 LinkedIn button: opens LinkedIn's own search, nothing is scraped.



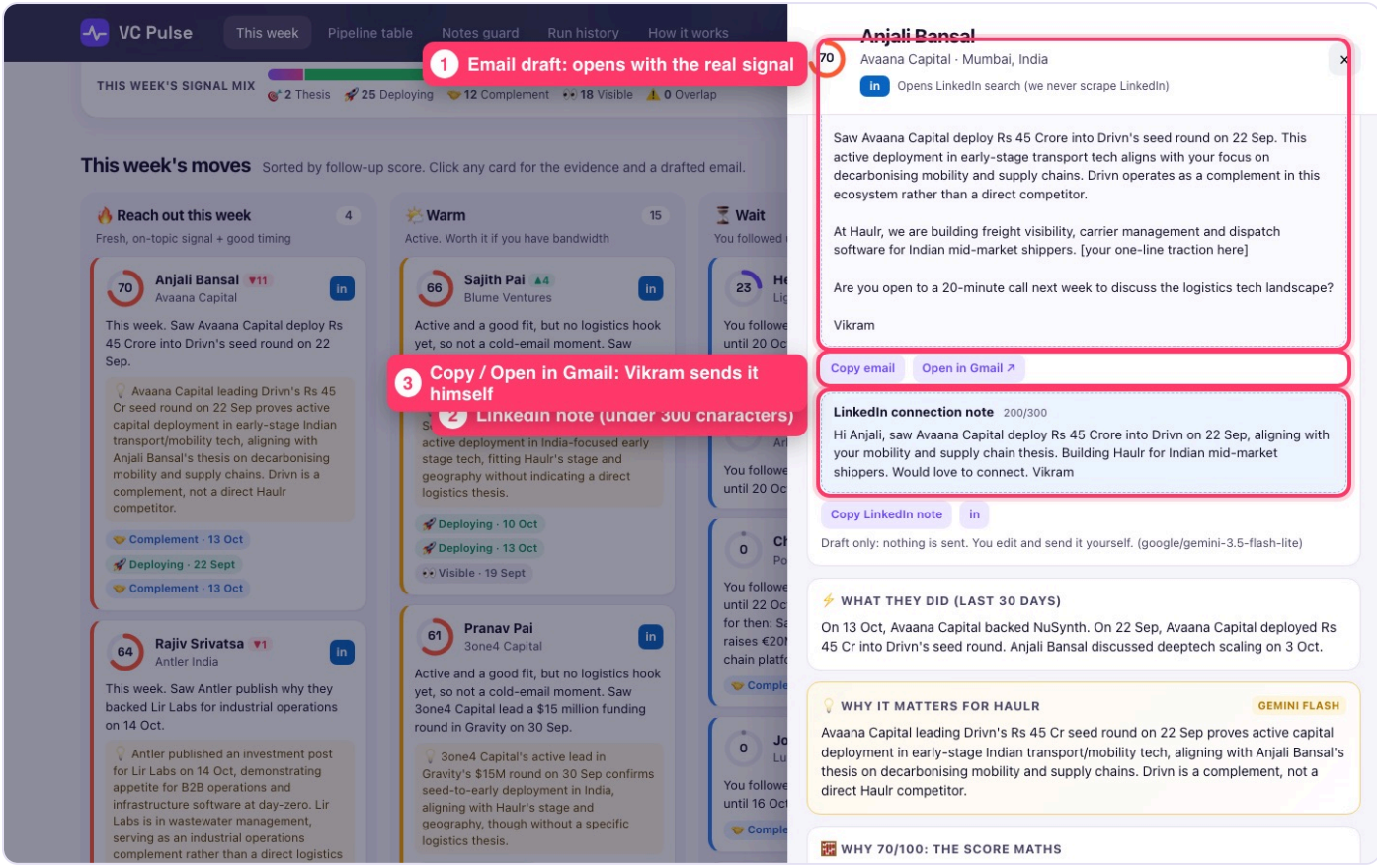
Click a card → the drawer. 1 The advice. 2 What they actually did in the last 30 days. 3 Why it matters for Haulr (Gemini Flash).



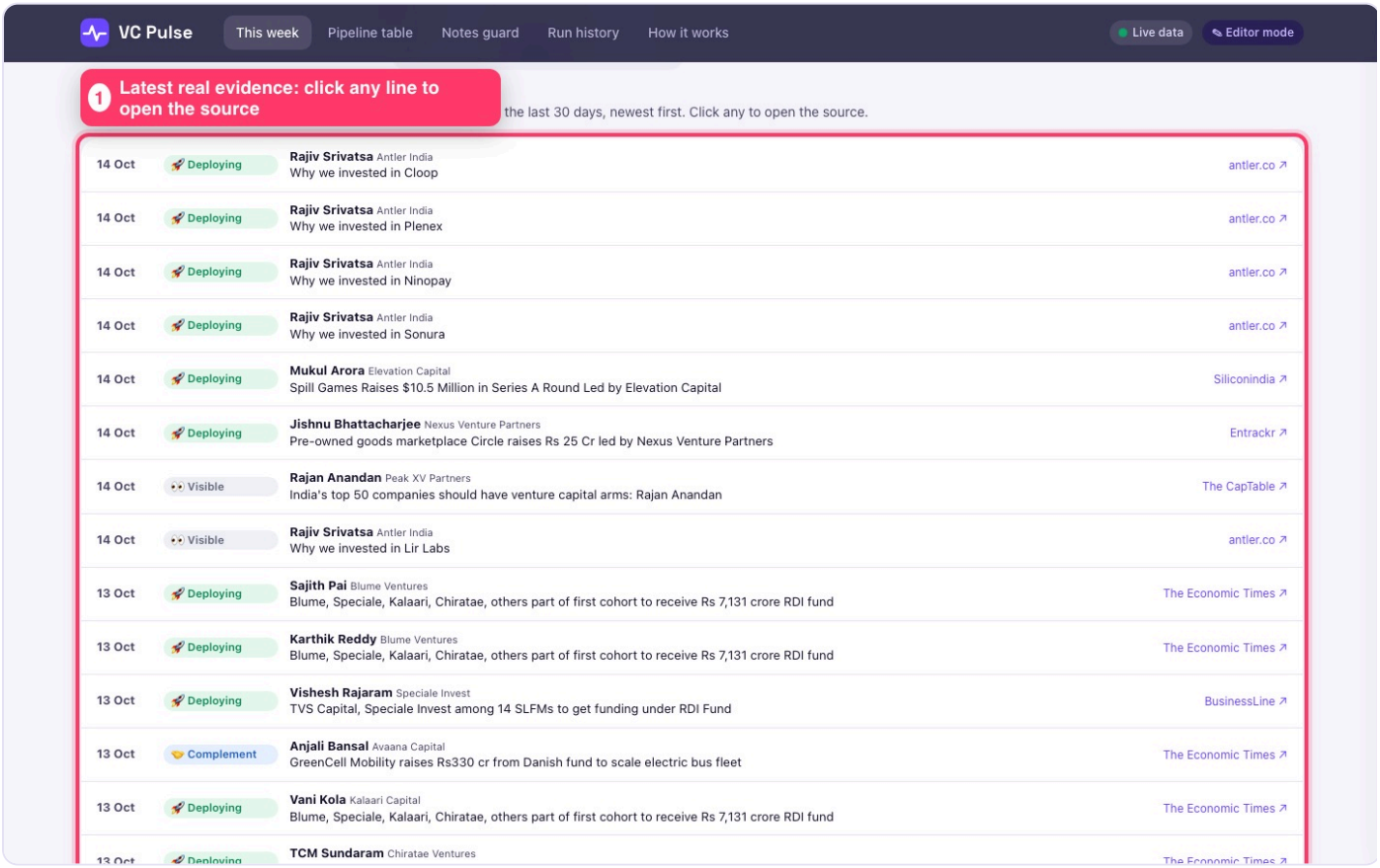
1 **The score maths, line by line**: signal strength, thesis fit, India fit, stage fit, timing. AI judges meaning; code does the arithmetic.



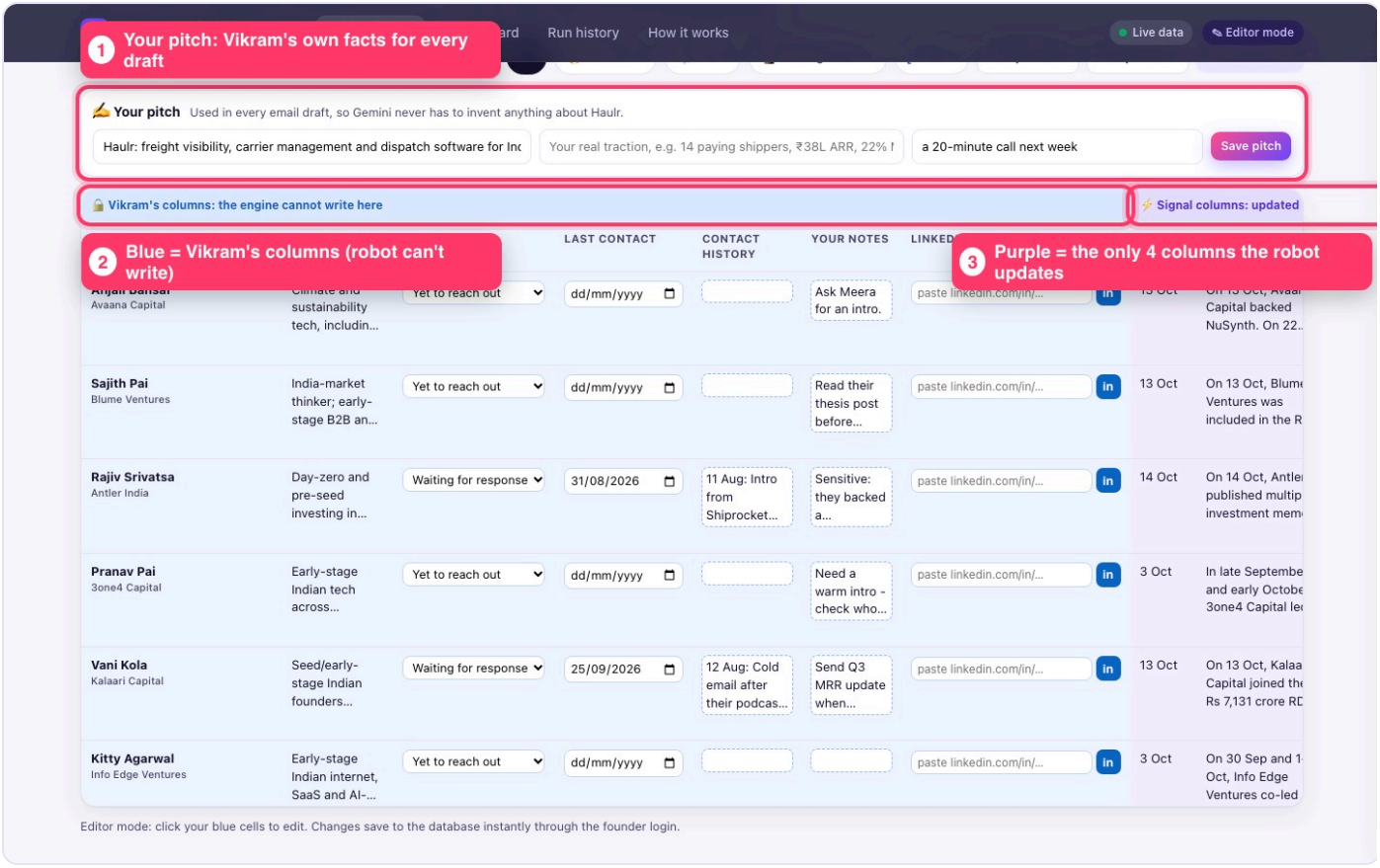
1 **Evidence**: every signal with its real link, date, source, meaning and points. "Same events, other outlets" are counted once; "Filtered out" shows what Gemini or our code rejected, so nothing is hidden.



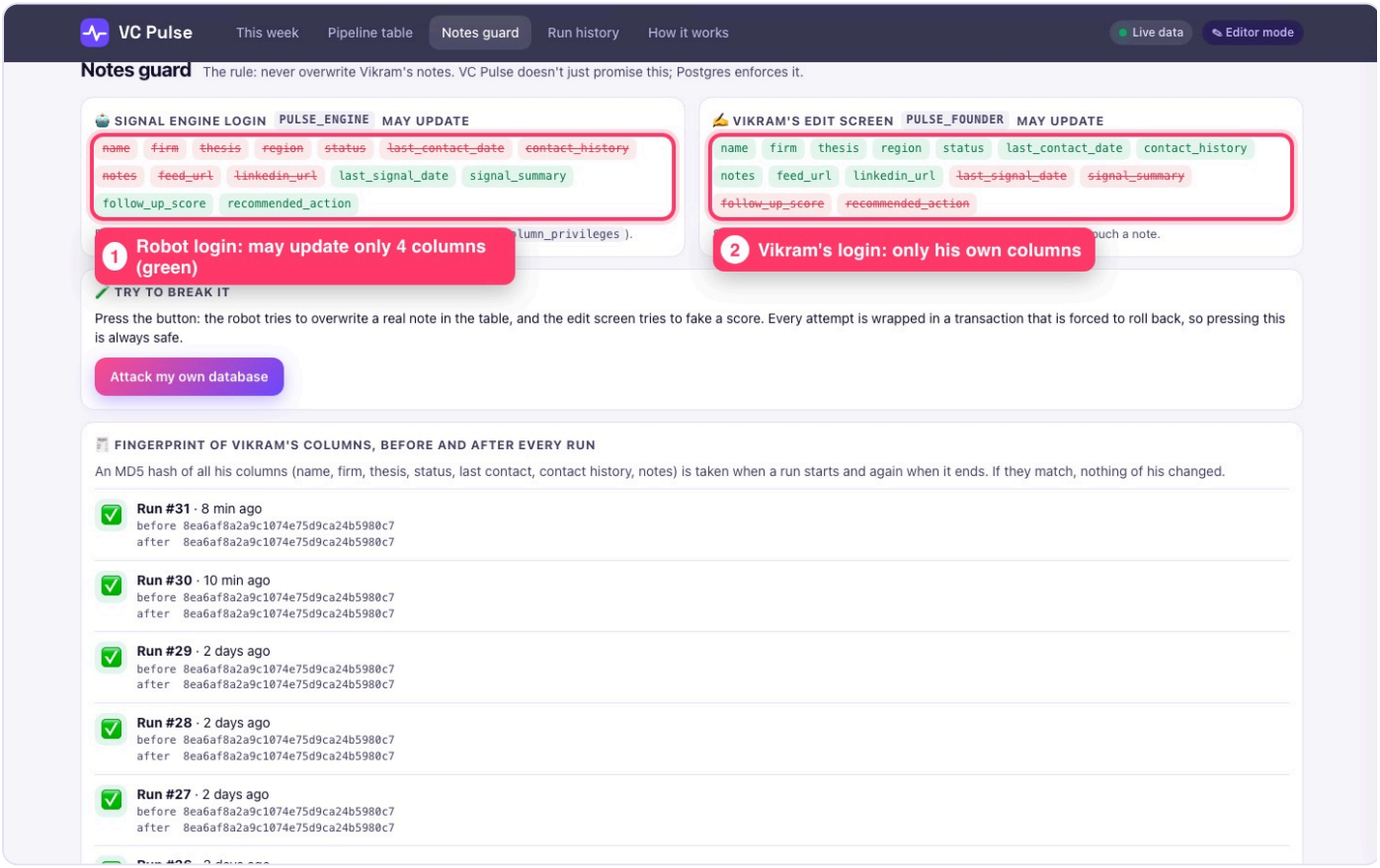
"Draft the email with Gemini". 1 An email that opens with the real signal, never invents traction (placeholder instead). 2 A LinkedIn note under 300 characters. 3 Copy / Open in Gmail / LinkedIn: Vikram sends it himself; nothing is sent automatically.



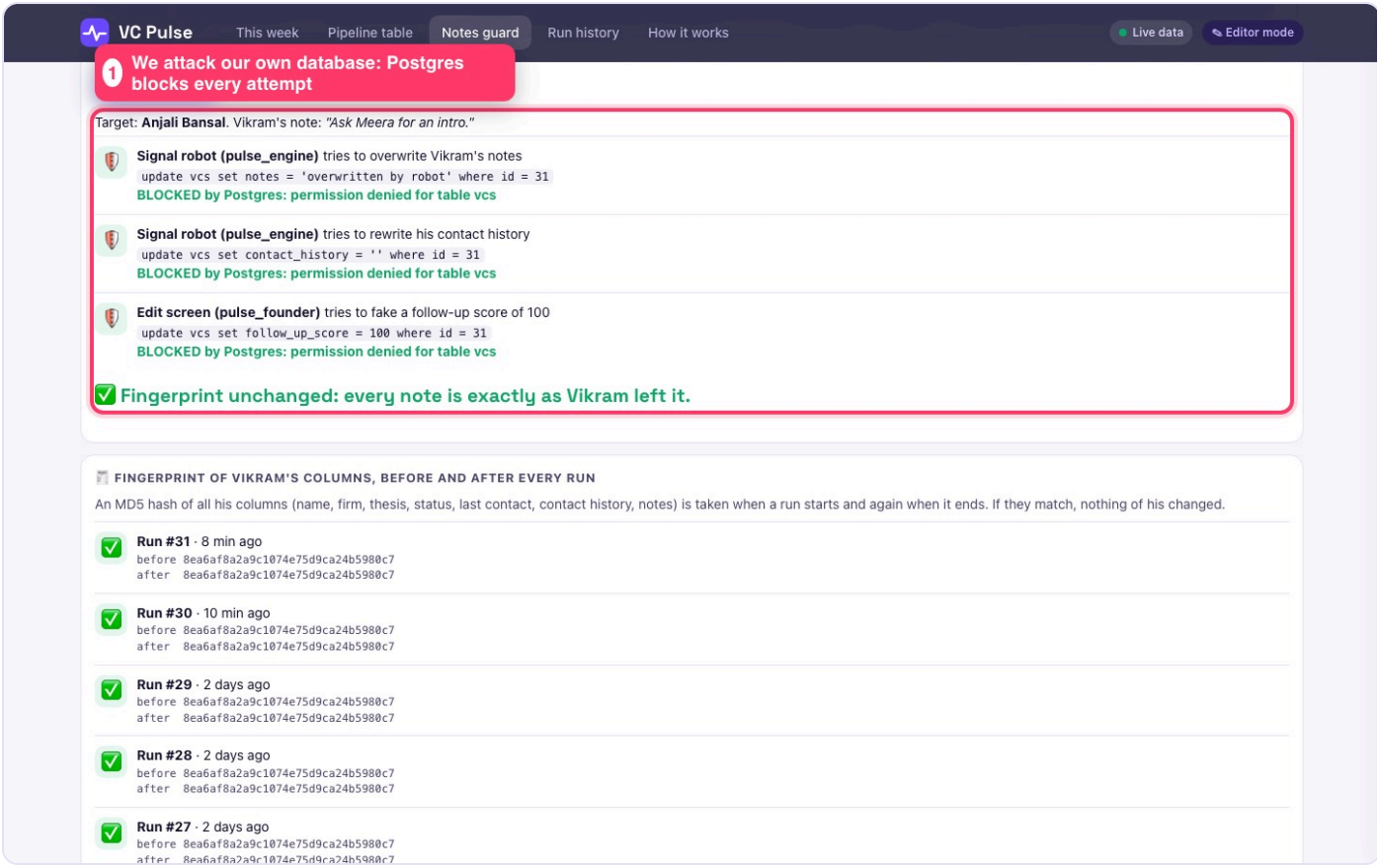
1 **Latest live evidence**: the newest real signals across all 50 investors. Click any line to open the original story (proof of requirement 1).



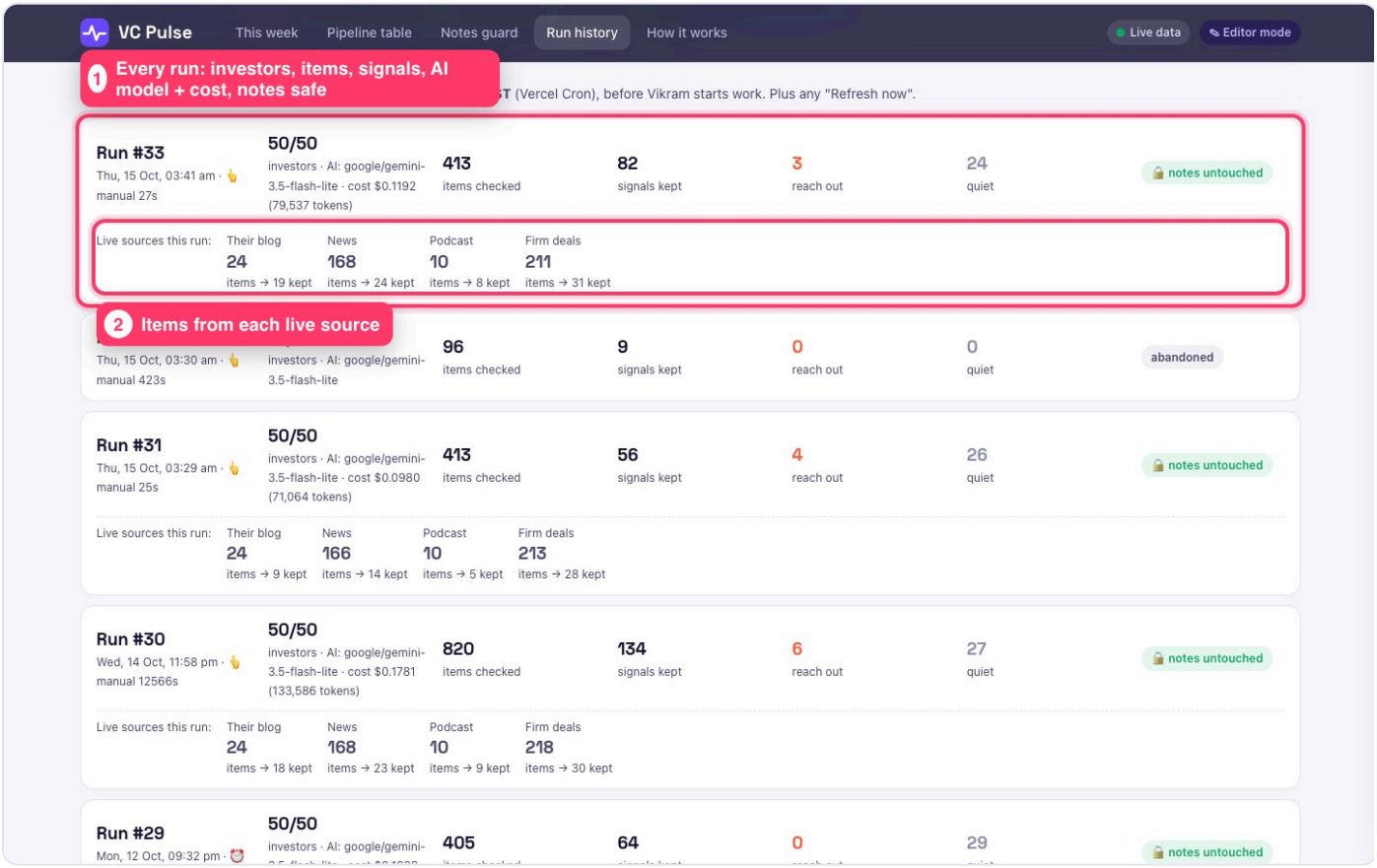
Pipeline table tab = the real database table. 1 "Your pitch": Vikram's own one-liner, traction and ask, used in every draft (editor only). 2 Blue = Vikram's columns (the robot can't write them). 3 Purple = the only 4 columns the robot updates. Search, filters, Export/Import CSV and "Add investor" are on top.



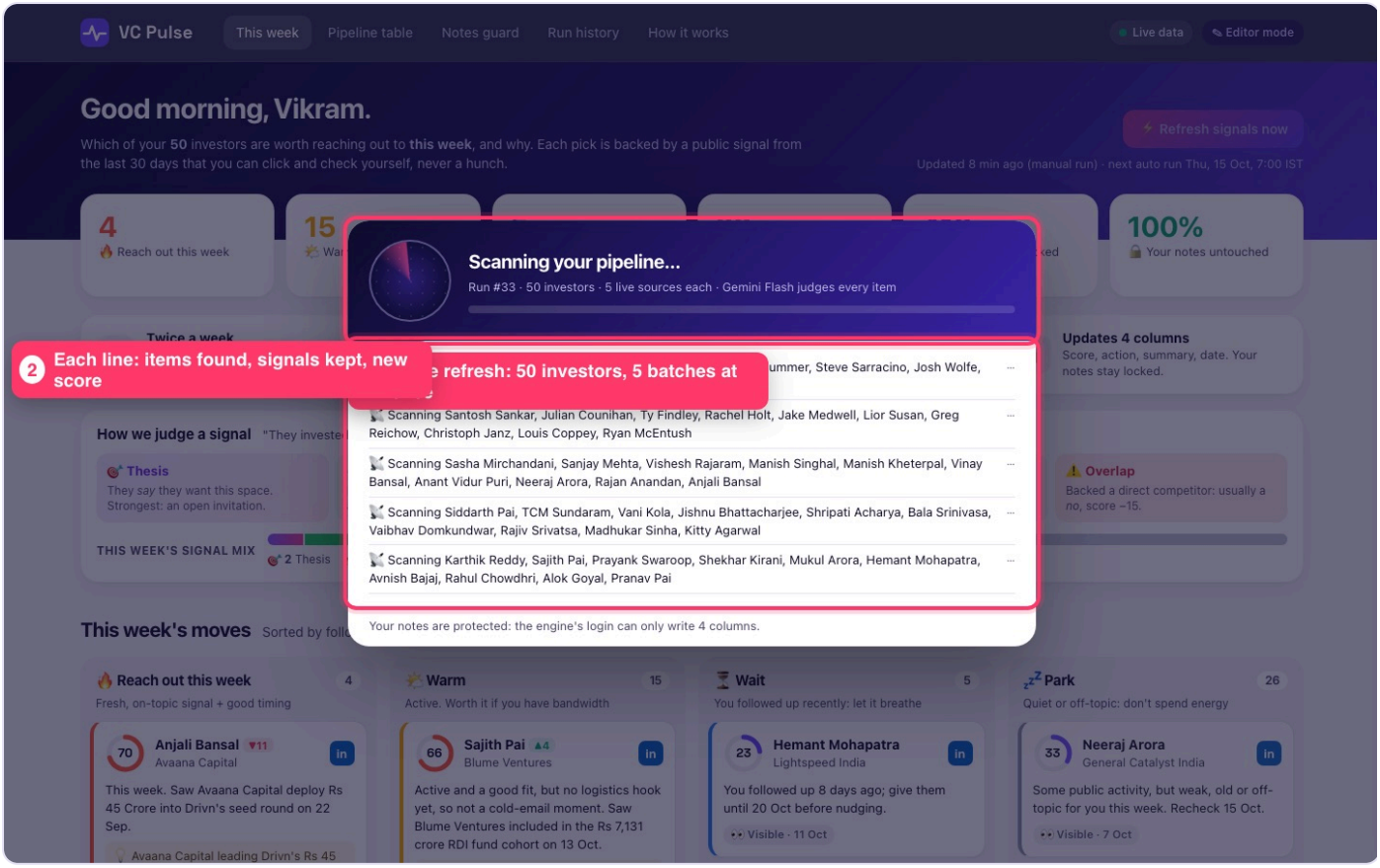
Notes guard tab. 1 The robot's database login: green = allowed (only the 4 signal columns), red = blocked. 2 Vikram's login: only his own columns. Read live from Postgres itself.



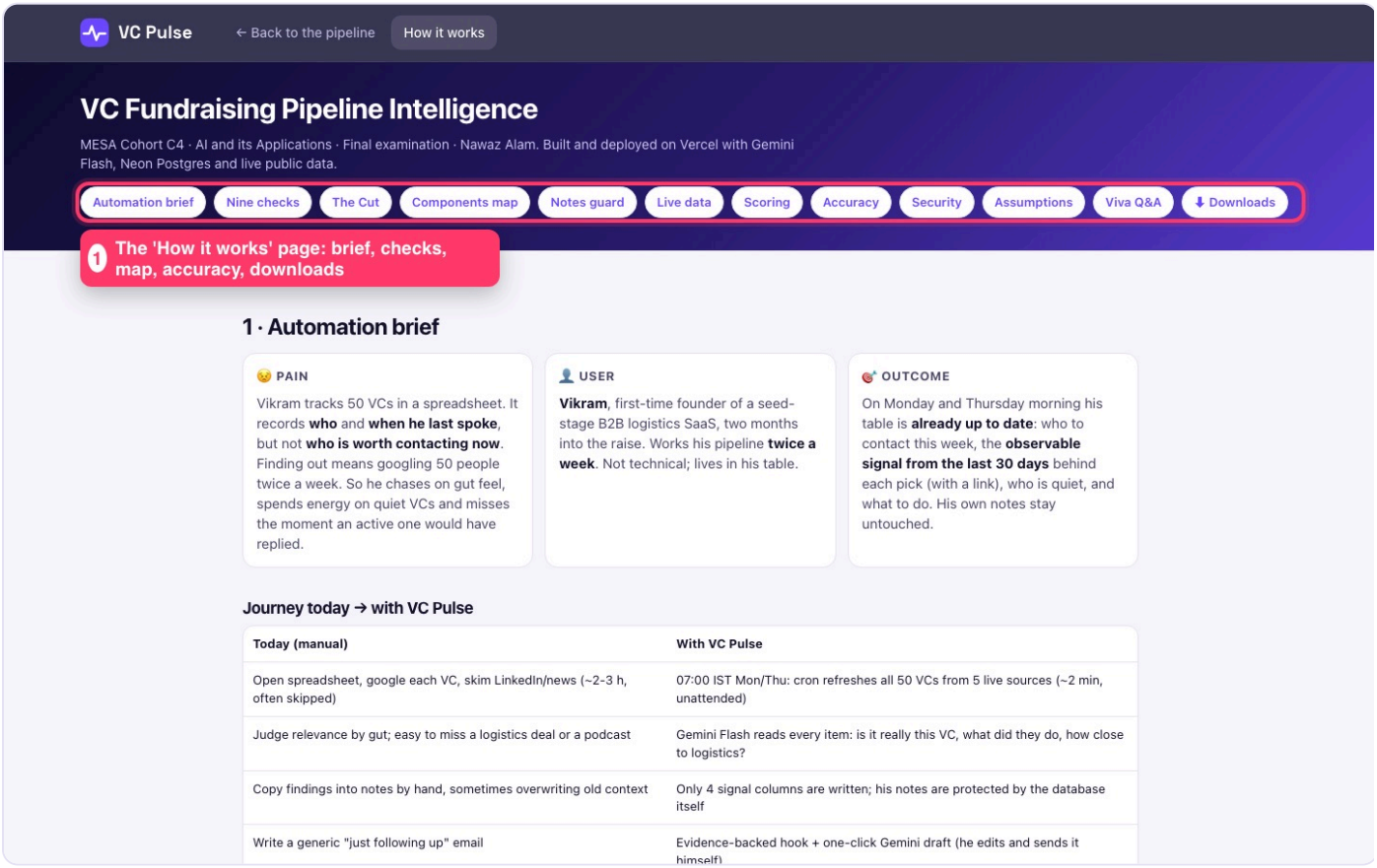
1 **"Attack my own database"**: the robot tries to overwrite a note and the edit screen tries to fake a score; Postgres answers "permission denied" each time, and the fingerprint of Vikram's notes stays identical. Safe to press any number of times.



Run history tab. 1 Each run: investors processed, items checked, signals kept, how many to reach out, the AI model and its cost (~$0.10), and "notes untouched". ⏰ = automatic run, 👆 = manual. 2 Items found per live source. A run left unfinished (e.g. a browser closed mid-refresh) is closed automatically and shown as "auto-closed".



A live refresh. 1 50 investors in 5 batches at once (~30-50 seconds). 2 Each line: items found → signals kept → new action and score (▲▼ = change).



1 **How it works page**: automation brief, Nine Checks, The Cut, Components Map, accuracy, security, assumptions, viva Q&A, and Downloads (map, checks, video, PDF).

Editor mode (top right, via Vikram's private link) unlocks editing his columns, saving his pitch, adding investors and refreshing any time. Without it the site is read-only and refresh has a 10-minute cool-down.

5 · How the score works

Step 1: what does each signal MEAN for Haulr? (the professor's key point)

"They invested in logistics before" is **not** a strong reason: a fund rarely backs a competitor of a company it already owns. So Gemini labels the *meaning*, and that decides the weight:

Meaning	Weight	Simple example
🎯 Thesis: they say they want this space	42	A partner's post: "why we're excited about freight software"
🚀 Deploying: writing new early cheques now	34	"Peak XV raises Surge seed cheques to $5M"
🤝 Complement: backed an adjacent, non-rival company	26	Backed an EV fleet company: same world, not a competitor
👀 Visible: active on other topics	10	A TV interview about AI: reachable, nothing more
⚠️ Overlap: backed a direct competitor	0 and −15	Backed another freight-visibility startup: usually a no

Points = weight × topic fit (logistics 1.0, adjacent 0.6, general 0.3) × who acted (the person 1.0, the firm 0.85) × freshness (full for 7 days, down to 40% at day 30). **Signal strength** = best + 40% of 2nd + 20% of 3rd (max 70). Then **fit**: thesis +3/+8/+15, India-based +10, seed +5, US-only −15, later-stage −8; and **timing**: emailed under 7 days ago −30, 7-13 days −10, fresh signal + good moment +10.

Step 2: the actions

Emailed under 14 days ago → **WAIT** · score ≥ 55 → **REACH OUT** (new investor, needs a logistics reason) or **NUDGE** (already pitched, any timely reason) · 35-54 → **WARM** · weak → **LOW** · nothing in 30 days → **PARK** (with how long they've been quiet).

Worked example (live, 15 Oct): Anjali Bansal, Avaana Capital = 75 → WARM

Best signals	🚀 Avaana invested ₹36.5 Cr in NuSynth (13 Oct, fit 2) = 23.1 pts; 🤝 a mobility/fleet item = 22.1; 🚀 Drivn ₹45 Cr seed (22 Sep) = 14.1
Strength	23.1 + 0.4×22.1 + 0.2×14.1 ≈ **35**
Fit and timing	thesis covers logistics/supply chain **+15** · India-based **+10** · seed stage **+5** · not contacted + timely **+10**
Total	**75**, but the strongest reason is a chemicals deal, not logistics → a cold email needs an on-topic hook → **WARM**: "Active and a good fit, but no logistics hook yet."

🎤 **Say this:** A high score means a good investor for us; the action also asks whether this week has a real reason to write. Both must be true for "reach out".

6 · Safety, reliability and cost

- **Notes guarantee (requirement 4):** two database logins. The robot (`pulse_engine`) may update only the 4 signal columns; Vikram's screen (`pulse_founder`) only his columns, including his saved LinkedIn link. A fingerprint (MD5) of his columns is taken before and after every run, and they always match.
- **Never wrong "quiet":** if a news source or the AI fails for someone, they keep last run's verdict; the run is marked "partial".
- **Never stuck:** one run at a time (database lock); a run left hanging is closed automatically after 6 minutes; the scheduled run always wins.

- **AI rules (course key sheet):** Gemini via the course OpenRouter key, Google models only, on Google's own servers; retries if busy; stops if credits run out.
- **Security:** secret keys only in Vercel settings; only real web links shown; strict browser security policy; editing needs Vikram's private key; public AI buttons are rate-limited.
- **LinkedIn, safely:** we never fetch LinkedIn; the "in" button just opens LinkedIn's search or his saved link.
- **Accuracy:** 100% precision, 85-89% recall, 93-95% accuracy on 57 hand-labelled items. **Cost:** ~$0.10 per run (Gemini 3.5 Flash-Lite: cheapest *and* most accurate in our test).

7 · Tech and AI, in simple words

Front end / back end	What you see in the browser (HTML, CSS, JavaScript) / the hidden code that does the work (Python on Vercel).
Vercel	Puts the website online; runs our Python code only when needed ("serverless"); its **Cron** is the twice-a-week alarm clock.
GitHub	Stores the code and its history; public = portfolio. Every update automatically redeploys the site.
Neon (Postgres)	The cloud database: Vikram's table, the evidence, the run log. It enforces who may write which column. (Same engine as Supabase; the brief's "Supabase table" works the same.)
API / API key	A way for programs to ask other programs for things / its secret password (kept in Vercel, never on GitHub).
RSS feed	A public list of a website's newest articles, made for apps to read (our live news source).
LLM · Gemini Flash	An AI that reads and writes language; Flash is Google's fast, cheap model family (exam rule). We use 3.5 Flash-Lite, with 3.7/3.8 Flash as backups.
OpenRouter	A gateway that forwards our AI requests; the course key; we force Google models on Google servers.
Prompt	The AI's job description and rules ("use only these headlines; never invent facts").
Structured output (JSON)	We force Gemini to answer in fixed boxes (who, what kind, meaning, fit), so code can read it reliably.
Temperature	The creativity dial: low (0.1) for judging, higher (0.6) for email drafts.
Hallucination / grounding	AI saying something untrue / making it answer only from real sources. Ours only labels items we fetched; code removes any company name not in the evidence.
Token · reasoning tokens	Pieces of words the AI reads and writes (that's what you pay for) · tokens it spends "thinking" (we set thinking to minimal: cheaper and just as accurate).
Precision · recall	Of what it flagged, how much was right (100%) · of what was really there, how much it found (85-89%).
Least privilege	Every part gets only the permissions it needs (the robot can write 4 columns, nothing else).
Hash / fingerprint	A short code computed from data; if anything changes, it changes. Used to prove notes were untouched.
Pipeline vs agent	We built a fixed pipeline (predictable, testable), not an autonomous AI agent (unpredictable, could touch the notes).

Your portfolio (all live)

App	Link	Key tech
VC Pulse (this final)	vc-pulse-lime.vercel.app	Vercel + Cron, Python, Neon, Gemini via OpenRouter, live RSS
Aangan Desk (Case 03, voice receptionist)	aangan-desk.vercel.app	Vaani voice AI, Neon, Gemini, Telegram, HubSpot, Cal.com
Kargo hiring (Case 02)	kargo-hiring-eight.vercel.app	Supabase, Gemini, CV scoring, Resend email
TripLock (group trip planner)	triplock-zeta.vercel.app	Vercel, Supabase

Skinstinct bot (Case 01) · Tonight's Pick (movie matchmaker)	GitHub (nawazalam4u)	Telegram bot · Supabase (paused to free a slot)

Common pattern across all of them: a real user problem → Nine Checks → Components Map → a Vercel app with a database, an AI step boxed in by code rules, and a human gate.

8 · The live demo: what to click and say (3 minutes)

1. **Open vc-pulse-lime.vercel.app.** "This is Vikram's week: 🔥 contact, 🌤️ warm, ⏳ wait, 💤 quiet. It refreshed itself this morning at 7."
2. **Point at "How we judge a signal".** "We don't trust 'they invested before'. We ask what each signal means: thesis, deploying, complement, visible, or overlap."
3. **Click the top card.** "Here's what they did, with real links; Gemini's reason why it matters; and the score maths line by line, done by code, not AI."
4. **Press "Draft the email with Gemini".** "It opens with the real signal and never invents facts. Vikram sends it himself."
5. **Notes guard → "Attack my own database".** "The robot tries to overwrite a note; Postgres refuses. The fingerprint is unchanged."
6. **Run history.** "Every run: items from each source, signals kept, cost about 10 cents, notes untouched."
7. **How it works page.** "Nine Checks, Components Map, accuracy, all downloadable."

If the Wi-Fi is slow: the dashboard still shows the last run's data; play the video instead. Don't press Refresh more than once; it takes ~40 seconds.

9 · Viva questions you're likely to get

About the problem and the Nine Checks

What problem did you solve?

A founder with 50 investors can't tell who to contact this week. VC Pulse reads live public signals twice a week, explains what they mean for his company, ranks investors, and updates his table, without touching his notes.

What are the Nine Checks and why do they matter?

A test of whether an AI tool deserves to be built: 3 kill switches (any no → don't build), 3 sizing checks (any no → build smaller), 3 boundary checks (any no → put a human there). Ours: all kill switches and sizing are yes; the boundaries put Vikram in charge of sending and judging.

Which check was hardest?

03, input available: LinkedIn has the richest data but scraping breaks its rules, so we chose official public feeds. And 06/07: we decided the AI never sends emails.

How do you measure impact (check 05)?

Time saved per session, % of emails with a real reason, reply rates, plus measured AI accuracy (100% precision) and cost per run logged.

What's "The Cut"?

What we deliberately didn't automate: sending emails, writing his notes, scraping LinkedIn, and letting AI set the score.

About the Components Map

Walk me through the map.

Trigger (cron or Refresh) → Input (Vikram's 50 investors) → Context (30 days of news, blogs, podcasts) → Processing (filters) → AI (Gemini judges meaning and explains why) → code checks and scores → Output (only 4 columns written, dashboard) → Vikram decides and sends.

Where exactly is the AI, and where is the human?

AI: the Gemini box judging each item and writing the "why" and drafts. Human: the gate at the end: Vikram reads, decides, sends; and only he edits his columns.

Why does code check the AI after Gemini?

AI can be wrong; code checks (name must appear, no memory facts, a deal isn't a thesis, sales aren't investing) keep precision at 100%.

What does "twice a week" look like in the map?

The Scheduler row: Vercel Cron, Monday and Thursday 07:00 IST, runs all 50 investors in about 30 seconds.

About the logic and the AI

"They invested in logistics before, so they're a strong lead." Agree?

No. A direct competitor makes them less likely (overlap, −15); an adjacent deal is only medium (complement). Strong signals: they say they want the space (thesis) or are writing new early cheques now (deploying), in India, at seed stage.

Why are most investors WARM, not REACH OUT?

Because a cold email needs a logistics reason. Many funds are active in other sectors: good fit, but no honest hook this week.

How do you stop the AI making things up?

It only labels real items we fetched (with links and dates), answers in fixed JSON, and code removes anything not supported by the evidence. Drafts leave a placeholder instead of inventing traction.

Why Gemini 3.5 Flash-Lite?

We tested three Flash models on 57 hand-checked items; Flash-Lite was cheapest and most accurate (100% precision). The exam requires Gemini Flash; it runs through the course's OpenRouter key.

How do you prove notes are never overwritten?

The database login the robot uses has no permission to write them; the Attack button shows "permission denied"; fingerprints match on every run.

What if something fails mid-run?

Failed investors keep last verdict, the run is marked partial, and a run left hanging is auto-closed after 6 minutes.

Why Neon if the brief says Supabase?

Both are Postgres; the same SQL works on either. Supabase's free slots were used by earlier projects.

How does LinkedIn work?

We never scrape it. The "in" button opens LinkedIn search, or a profile link Vikram saved in his own protected column.

What would you build next?

Gmail integration (with consent) to log replies and learn which signals predict replies; logins for many founders; reading full articles, not just headlines.

What did you learn?

Use AI for judgment on real evidence, box it in with rules and code checks, measure it, and let the database enforce the rules that matter most.

Last tip: if you don't know an answer, say what the system does and point to it on screen: "Let me show you": open the drawer, the score maths, or the Notes guard. Showing beats explaining.